# JOÃO RODRIGUES

PALO ALTO, CA | JOAOR.EU | CV@JOAOR.EU

## Summary

I am a senior computational biologist with a proven track record bridging chemistry, biology, and computer science. My work has resulted in 46 peer-reviewed publications and several widely adopted open-source scientific software packages, totaling over 9,500 citations. At Schrödinger, I am a senior member of the Protein Structure Prediction group, contributing to core induced-fit docking methods and leading a team developing new methods for cryo-EM structure optimization and macrocycle modeling. In addition, I head the scientific development of the Protein Preparation Workflow, a cornerstone of the Schrödinger suite and its most-used tool. Alongside research, I enjoy fostering communities in computational biology, having helped organize several courses and workshops, and am regularly invited as a speaker or panelist to national and international conferences.

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## Experience

**Schrödinger Inc** | Senior Principal Scientist (promoted through Sr. Scientist) 2021 - Present

- Daily supervision of a team of scientists and engineers working on several core products of the Schrödinger computational chemistry software suite.
- Spearheaded the creation of protocols, skills, and tests for Bunsen, an agentic AI co-scientist, in tandem with the Machine Learning team.
- Designed and led a benchmark study on the real-world applicability of co-folding models for drug discovery, accepted for publication in npj Drug Discovery.
- Led scientific and engineering efforts on key macrocycle modeling products, resulting in improved scientific performance, a new docking method, renewed customer interest, several poster and oral presentations, and a publication in J. Med.Chem.
- Led the implementation of several improvements and new features into the backend and frontend of the Protein Preparation Workflow, logging >250k executions per year, in collaboration with the UX/UI and scientific infrastructure teams.
- Led the development of an integrated suite of methods and tests to optimize and validate structures of relevant drug discovery targets against low-resolution cryo-EM data, actively used both by the Structure Therapeutics Group at Schrödinger and external customers.
- As an individual contributor, eliminated critical performance bottlenecks in flexible ligand docking approaches (IFD-MD) and implemented and validated new algorithms to extend their applicability to membrane proteins such as GPCRs.

**Biopython Project** | Maintainer and Core Developer 2010 - Present

- Maintainer of the Bio.PDB module for structural bioinformatics.
- Implemented solvent-accessible surface area calculation and structure alignment code.
- Oversaw, reviewed, and shepherded several individual contributions over the years.

**Stanford University, Levitt Lab** | Postdoctoral Fellow 2016 - 2020

- Developed simulation methods to study the structure, dynamics, and interactions of membrane proteins, resulting in publications in Cell, Nat. Chem. Bio., and PNAS.
- Designed and built a framework based on structural homology and molecular mechanics to predict cross-species transmission of SARS-CoV-2, resulting in 3 highly-cited publications and several national and international collaborations.
- Contributed several bug fixes and new features to OpenMM, a widely-used open-source engine for molecular dynamics simulations.
- Co-developed an immersive virtual reality approach to protein docking using Unity.

**Utrecht University, Bonvin Lab** | Doctoral and Postdoctoral Researcher 2010 - 2016

- Core contributor to HADDOCK, a protein docking software with over 90,000 users across 190 countries, and consistently ranked among the best in CASP/CAPRI assessments.
- Co-developed coevolution-based methods for protein quaternary structure prediction, a precursor to AlphaFold-based approaches.
- Created a widely-used open-source toolkit to manipulate atomic structures (pdb-tools).
- Created materials for undergraduate and graduate courses on structural bioinformatics.

## Selected Publications

# indicates corresponding author, * indicates co-first author

• Zhang, L, Friesner, RA, Miller, EB, and **Rodrigues, JP**# (2026). Generalization and Usability of Co-Folded GPCR–Ligand Complexes: A Physics-Guided Assessment. Preprint

• Robson-Tull, J, and **Rodrigues, JP**# (2026). Accurate Physics-Based Flexible Docking of Macrocyclic Ligands. J. Med. Chem.

• Miller, EB, Hwang, H, Shelley, M, Placzek, A, **Rodrigues, JP**, et al. (2024). Enabling structure-based drug discovery utilizing predicted models. Cell.

• Eastman, P, Galvelis, R, Peláez, RP, Abreu, CRA, Farr, SE, et al. (2024). OpenMM 8: Molecular Dynamics Simulation with Machine Learning Potentials. J. Phys. Chem. B.

• **Rodrigues, JP**#, Barrera-Vilarmau, S, M. C. Teixeira, J, Sorokina, M, Seckel, E, et al. (2020). Insights on cross-species transmission of SARS-CoV-2 from structural modeling. PLoS Comput Biol.

• Van Zundert, GCP, **Rodrigues, JP***, Trellet, M, Schmitz, C, Kastritis, PL, et al. (2016). The HADDOCK2.2 Web Server: User-Friendly Integrative Modeling of Biomolecular Complexes. Journal of Molecular Biology.

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## Selected Community Engagement

**Invited Speaker**, Pittsburgh Diffraction Conference 2026

**Invited Panelist**, Biophysical Society Meeting 2026

**Invited Panelist**, West Coast Structural Biology Workshop 2025

**Invited Keynote**, MICROBIOTEC'25, Azores (declined) 2025

**Invited Panelist**, Careers in Comp Chem, New York Area Group for Informatics and Modeling 2024

**Scientific Committee**, HIBIT Conference 2019-2025

**Co-organizer**, EMBO Practical Course on "Integrative Modeling of Biomolecular Interactions" 2016-2020

**Invited Lecturer**, Summer School "Exploring Nature's Molecular Machines", at Uni. Utrecht 2014

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## Awards and Grants

**The Molecular Sciences Software Institute (MOLSSI)** - Seed Fellowship ($25,000) 2020

**ISMB/ECCB - RCSB PDB** Poster Prize - Outstanding Student Paper 2019

**Niels-Stensen Fellowship** for Postdoctoral Studies ($44,000) 2016

Stanford School of Medicine - **Dean's Postdoctoral Fellowship** ($25,000) 2016

**Google Summer of Code Fellowship**, Open Bioinformatics Foundation ($5,000) 2010

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## Education

**Ph.D. in Computational Structural Biology** 2010 - 2014
*Bonvin Lab, Universiteit Utrecht, The Netherlands*
Thesis: Computational Structural Biology of Macromolecular Interactions

**M.Sc. in Biomolecular Sciences** 2008 - 2010
*Universiteit Utrecht, The Netherlands*
Thesis: Predicting the Specificity of E2/E3 Complexes in the Ubiquitination Pathway

**B.Sc. in Biochemistry** 2004 - 2008
*Universidade de Coimbra, Portugal*
Thesis: Concept-based Query Expansion for Gene Association Prediction from Literature Data

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## Skills

**Tech Stack** Python, Cython, Bash, Git, SQL (Basic), JavaScript (Basic)

**Databases** RCSB, NCBI, ChEMBL, BLAST, KEGG, Interpro, GPCRdb, OPM

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Github github.com/joaorodrigues

Google Scholar scholar.google.com/citations?user=UeXRdzQAAAAJ